Wilson-Cowan Model

In computational neuroscience, the **Wilson-Cowan model** (Hugh R. Wilson & Jack D. Cowan) describes the dynamics of interactions between populations of very simple excitatory and inhibitory model neurons. It is a mean-field model that tracks the **proportion of active neurons** in each population over time.

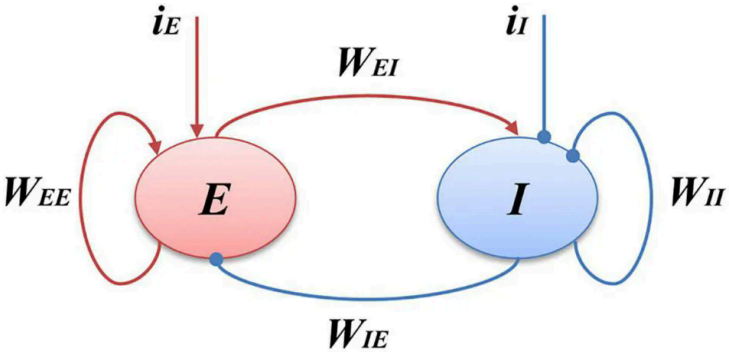

Figure 1: Schematic of the Wilson-Cowan model[1]

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) \tag{1}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{2}$$

where the activation function F is a sigmoid:

$$F(x; a, \theta) = \frac{1}{1 + e^{-a(x-\theta)}} - \frac{1}{1 + e^{a\theta}} \tag{3}$$

The correction term $-\frac{1}{1+e^{a\theta}}$ ensures $F(0) = 0$ (no input yields no activation).

Remark: We use the simplified Wilson-Cowan equations without the refractory term $(1 - r_E)$, following the formulation in Neuromatch Academy.

0.1 Variables

- $r_E(t)$ — average firing rate (or proportion of active neurons) of the **excitatory** population at time t. Ranges from 0 (silent) to 1 (maximally active).
- $r_I(t)$ — average firing rate of the **inhibitory** population at time t.

0.2 Parameters

0.2.1 Time Constants

- τ_E — time constant of the excitatory population.
- τ_I — time constant of the inhibitory population.

[1] E represents the excitatory populations, I stands for the inhibitory populations, the red arrows indicate the excitatory projections and blue arrows show the inhibitory projections, i_I is the external inputs of the I populations, and i_E is the external inputs of the E populations. Li et al. (2022)

0.2.2 Sigmoid Parameters

- a_E, a_I — gain (steepness) of the activation function for E and I populations. Higher gain means a sharper threshold-like response.
- θ_E, θ_I — threshold of the activation function. The input level at which the population reaches half-maximal activation.

0.2.3 Connection Weights

- w_{EE} — excitatory → excitatory coupling strength (recurrent excitation).
- w_{EI} — inhibitory → excitatory coupling strength (how much I suppresses E).
- w_{IE} — excitatory → inhibitory coupling strength (how much E drives I).
- w_{II} — inhibitory → inhibitory coupling strength (recurrent inhibition).

0.2.4 External Inputs

- I_E^{ext} — external input current to the excitatory population (primary control parameter for bifurcation analysis).
- I_I^{ext} — external input current to the inhibitory population (typically fixed at 0).

0.2.5 Parameter Values

Parameter	Value	Description
τ_E	1.0 ms	Average synaptic time constant for excitatory population
τ_I	2.0 ms	Average synaptic time constant for inhibitory population
a_E	1.2	Excitatory gain
a_I	1.0	Inhibitory gain
θ_E	2.8	Excitatory threshold
θ_I	4.0	Inhibitory threshold
w_{EE}	9.0	Strength of self-excitatory feedback
w_{EI}	4.0	Strength of connections between E–I
w_{IE}	13.0	Strength of connections between I–E
w_{II}	11.0	Strength of self-inhibitory feedback
I_E^{ext}	0.0	External input to excitatory population (sweep this)
I_I^{ext}	0.0	External input to inhibitory population

Table 1: Default parameter values from the Neuromatch Academy computational neuroscience curriculum.

Remark: The parameters play an important role in the system, please read the *Periodic solution (Oscillations)* part in details.

1 Nullcline Analysis

1.1 E-nullcline

Let $\dot{r}_E = 0$, then

$$\frac{1}{\tau_E}(-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}})) = 0 \tag{4}$$

$$-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = 0 \tag{5}$$

$$F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = r_E \tag{6}$$

Then we use the inverse of F (namely the logit function):

$$F^{-1}(y; a, \theta) = \theta + \frac{1}{a} \ln \frac{y + c}{1 - y - c}, \quad c = \frac{1}{1 + e^{a\theta}} \tag{7}$$

Applying F_E^{-1} to both sides:

$$F_E^{-1}(r_E) = w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}} \tag{8}$$

Solving for r_I:

$$r_I = \frac{1}{w_{EI}}(w_{EE} \cdot r_E - F_E^{-1}(r_E) + I_E^{\text{ext}}) \tag{9}$$

1.2 I-nullcline

Analogously, let $\dot{r}_I = 0$, we have

$$r_I = F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{10}$$

$$r_E = \frac{1}{w_{IE}}(w_{II} \cdot r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \tag{11}$$

1.3 Phase plane

After substituting the parameters, we obtain the figure

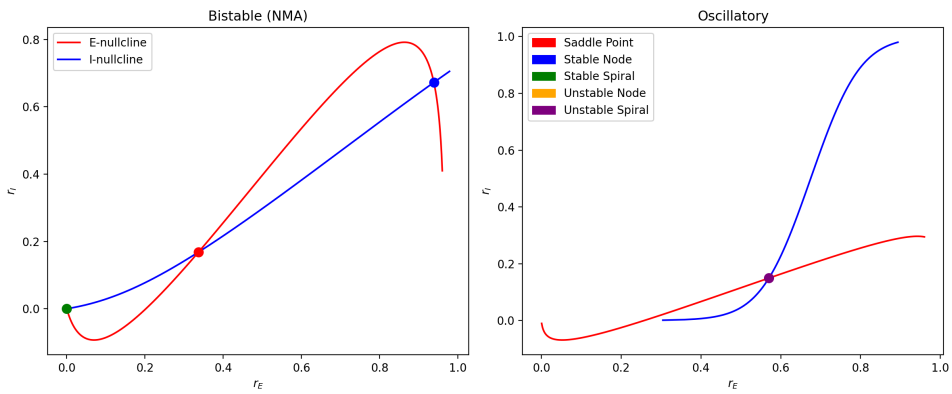

Figure 2: Phase plane analysis

Inserting point $(0.1, 0.5)$, it's easy to check the arrow direction. We omit here.

Remark: The domains are $r_E \in (0, 0.9664)$ and $r_I \in (0, 0.9820)$, as the logarithm requires its argument to be positive.

2 Qualitative Analysis

Note that in the previous section, we obtain the phase plane with 3 equilibrium points. Now we use the Jacobian matrix to determine its state. Recall the Jacobian matrix:

$$J = \begin{pmatrix} \frac{\partial \dot{r}_E}{\partial r_E} & \frac{\partial \dot{r}_E}{\partial r_I} \\ \frac{\partial \dot{r}_I}{\partial r_E} & \frac{\partial \dot{r}_I}{\partial r_I} \end{pmatrix} \tag{12}$$

We solve the partial derivative and obtain

$$J = \begin{pmatrix} \frac{1}{\tau_E}(-1 + w_{EE} \cdot F'_E) & \frac{1}{\tau_E}(-w_{EI} \cdot F'_E) \\ \frac{1}{\tau_I}(w_{IE} \cdot F'_I) & \frac{1}{\tau_I}(-1 - w_{II} \cdot F'_I) \end{pmatrix} \tag{13}$$

in which

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c), \quad c = \frac{1}{1 + e^{a\theta}} \tag{14}$$

Note that F' is evaluated at the sigmoid input, not at r_E or r_I directly:

$$u_E = w_{EE} r_E - w_{EI} r_I + I_E^{\text{ext}}, \quad u_I = w_{IE} r_E - w_{II} r_I + I_I^{\text{ext}} \tag{15}$$

We classify each fixed point using the trace $\tau = \text{tr}(J)$, determinant $\Delta = \det(J)$, and discriminant $\tau^2 - 4\Delta$ of the Jacobian:

- $\Delta < 0 \Rightarrow$ saddle point
- $\Delta > 0, \tau < 0 \Rightarrow$ stable (node if $\tau^2 - 4\Delta > 0$, spiral if < 0)
- $\Delta > 0, \tau > 0 \Rightarrow$ unstable

For our 3 fixed points:

Fixed Point	τ	Δ	$\tau^2 - 4\Delta$	**Classification**
$(0, 0)$	-1.247	0.406	-0.069	Stable Spiral
$(\approx 0.34, 0.17)$	—	-0.941	—	Saddle Point
$(\approx 0.94, 0.68)$	-2.403	1.381	0.250	Stable Node

Table 2: Fixed point classification for default parameters ($I_E^{\text{ext}} = 0$).

The system with default parameters ($I_E^{\text{ext}} = 0$) is therefore **bistable**. (In a dynamical system, **bistability** means the system has two stable equilibrium states a bistable structure can be resting in either of two states.)

3 Periodic Solutions (Oscillations)

Recall the **Poincaré-Bendixson theorem** states that for a two-dimensional system, for which we know that the solutions will stay in **a bounded area** after some finite time and

there is **no fixed point**, there must exist a periodic solution to which the trajectories converge.

We sweep the I_E^{ext} until we obtain only one fixed point which is unstable.

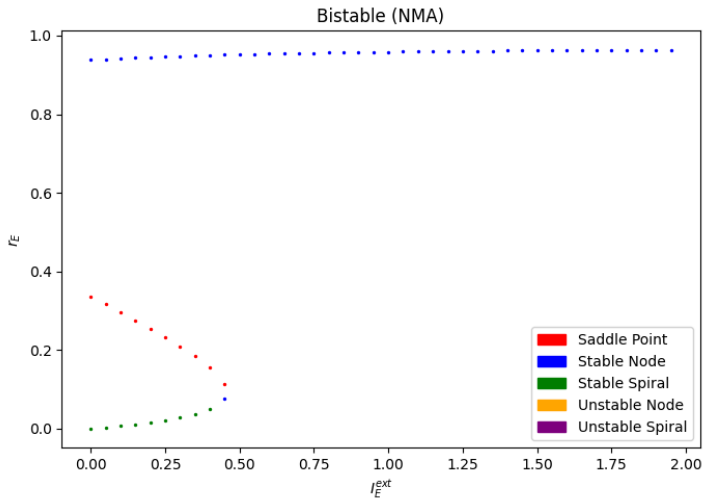

Figure 3: Stable fixed point output after sweeping I_E^{ext}

Note that after sweeping I_E^{ext}, we only get the stable 1 fixed point, why?

Comparing with the parameters used in Li et al. (2022) (Table 1, based on Wilson and Cowan (1972) and Jadi and Sejnowski (2014)) for gamma oscillation research:

Parameter	**NMA (bistability)**	**Table 1 (oscillation)**
w_{EE}	9	16
w_{EI}	4	26
w_{IE}	13	20
w_{II}	11	1
τ_E (ms)	1	20
τ_I (ms)	2	10
θ_E, θ_I	2.8, 4.0	5, 20
$I_E^{\text{ext}}, I_I^{\text{ext}}$	0, 0	2, 7

Table 3: Parameter comparison: bistable vs. oscillatory regime.

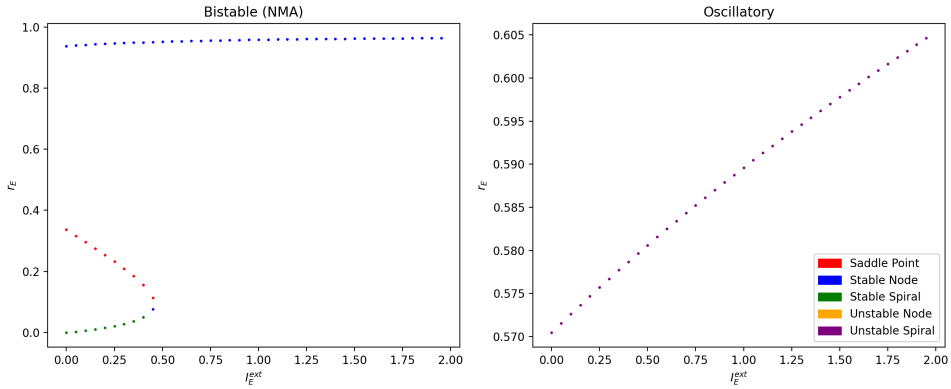

Figure 4: Output comparison between NMA (bistable) and Li et al. (oscillatory) parameter regimes.

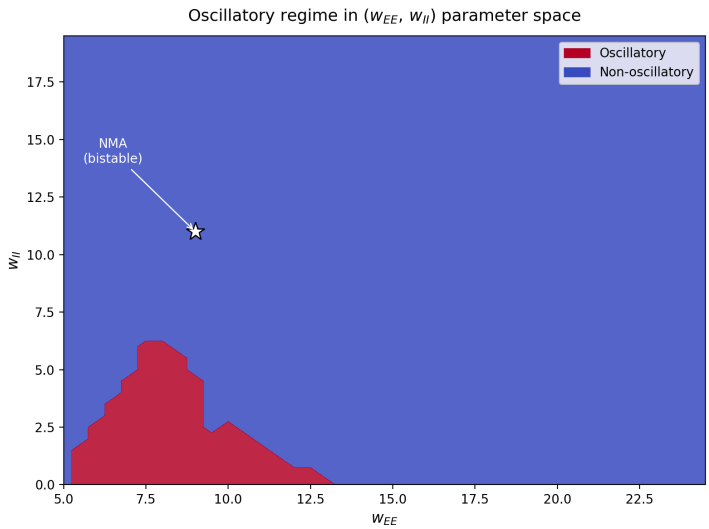

Figure 5: Oscillatory regime in (Wee, WI) parameter space for NMA parameters

3.1 Conditions for Oscillation

A Hopf bifurcation requires a fixed point where the Jacobian has both positive trace (instability) and positive determinant (complex eigenvalues):

$$\tau = \frac{1}{\tau_E}(-1 + w_{EE}F_E') + \frac{1}{\tau_I}(-1 - w_{II}F_I') \tag{16}$$

$$\Delta = \frac{1}{\tau_E\tau_I}[1 + w_{II}F_I' - w_{EE}F_E' + (w_{EI}w_{IE} - w_{EE}w_{II})F_E'F_I'] \tag{17}$$

- $\tau > 0$: In order to let $\frac{1}{\tau_E}(-1 + w_{EE}F_E') + \frac{1}{\tau_I}(-1 - w_{II}F_I') > 0$, we need

$$w_{EE}F_E' > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') \tag{18}$$

  We require $w_{EE}F_E'$ to be large enough. The sigmoid derivative F_E' is maximised when $u_E = w_{EE}r_E - w_{EI}r_I + I_E^{\text{ext}}$ lies near the midpoint θ_E. Consider two cases, if w_{EE} is too small, then u_E stays below θ_E and F_E' is small. If w_{EE} is too large, strong positive feedback pushes the fixed point past θ_E into saturation, and F_E' is again small. Therefore, w_{EE} must lie in a sweet spot to keep the inequality holding. Ideally it should close to θ_E and the F_E' should near the maximum $\frac{1}{4}a_E$.

- $\Delta > 0$: In order to let

$$\frac{1}{\tau_E \tau_I}[1 + w_{II}F_I' - w_{EE}F_E' + (w_{EI}w_{IE} - w_{EE}w_{II})F_E'F_I'] > 0 \tag{19}$$

we require

$$1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I' > w_{EE}F_E' + w_{EE}w_{II}F_E'F_I' \tag{20}$$

$$1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I' > w_{EE}F_E'(1 + w_{II}F_I') \tag{21}$$

From inequality 21, we note that w_{EE} has an upper bound, as we previously derived. From inequality 20, we arrive at the most critical condition: in order to let the inequality hold, it must follow $w_{EI}w_{IE} > w_{EE}w_{II}$. To prove this, let's assume otherwise. Then we require $1 + w_{II}F_I' > w_{EE}F_E'$[2]. But we want $\tau > 0$ as well, combined with inequality 18, we obtain

$$1 + w_{II}F_I' > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') > \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') \tag{22}$$

Since $1 + w_{II}F_I'$ is strictly positive, dividing both sides by it yields:

$$1 > \frac{\tau_E}{\tau_I} \Rightarrow \tau_E < \tau_I \tag{23}$$

However, from Keeley et al. (2019)

> …weakly modulated network gamma is found when the firing rate of the inhibitory population is recruited faster than that of the excitatory population, and strongly synchronized gamma oscillations are seen for the reverse. In both frameworks, network oscillations arise when a steady state loses stability via Hopf bifurcation as drive to the excitatory population increases. The small-amplitude oscillation that emerges from the bifurcation persists over a substantial range of values of the drive in the weakly modulated regime, but for the strongly synchronous case the oscillation amplitude rises sharply to become a strongly synchronous E-I rhythm as drive increases.

It shows that the Oscillation may occur when $\tau_E < \tau_I$. However, this regime corresponds to weakly modulated rhythms rather than typical Gamma oscillations, which are the primary focus of this study. Therefore, we exclude this scenario. Hence it's a contradiction, we need $w_{EI}w_{IE} - w_{EE}w_{II} > 0$ to keep the $\Delta > 0$.

We conclude our observation as 3 conditions:
- Condition 1: w_{EE} has lower bound.
- Condition 2: w_{EE} has upper bound[3].
- Condition 3: $w_{EI}w_{IE} > w_{EE}w_{II}$.

[2]And the difference of it must be greater than the difference of $w_{EE}w_{II} - w_{EI}w_{IE}$.

[3]However, which upper bound is tighter depends on the specific parameters. Mostly depends on cross-term product $w_{EI}w_{IE}$. It decides if the system can survive a large w_{EE} without the determinant becoming negative. In rare cases, w_{II} and the ratio $\frac{\tau_E}{\tau_I}$ also have a small effect. This is because w_{II} appears on both sides of the determinant inequality, and $\frac{\tau_E}{\tau_I}$ changes how easily the trace drops below zero during saturation. Overall, looking at the basic network structure, it is correct to say that the strength of $w_{EI}w_{IE}$ controls this outcome.

[4]To verify it, use our condition test module.

3.2 Verification[4]

Numerical evaluation at selected points in the (w_{EE}, w_{II}) parameter space confirms all three conditions. At oscillatory points (e.g. $w_{EE} = 11$, $w_{II} = 0$, $I_E^{\text{ext}} = 0.50$), the system satisfies all three conditions and exhibits a single Unstable Spiral.

Points that violate individual conditions fail to oscillate:

- $(w_{EE} = 11, w_{II} = 5)$: The self-feedback product $(11 \times 5 = 55)$ exceeds the cross-term product $(w_{EI}w_{IE} = 52)$ — Condition 3 fails.
- $(w_{EE} = 14, w_{II} = 2)$: The cross-term is valid $(52 > 28)$. However, w_{EE} is too large, which amplifies the negative term $-w_{EE}F'_E$. This overpowers the cross-term and forces the determinant below zero $(\Delta < 0)$, collapsing the fixed point into a Saddle — Condition 2 fails.
- $(w_{EE} = 5, w_{II} = 0)$: The cross-term and determinant are valid, but $w_{EE}F'_E$ is insufficient to overcome the decay — Condition 1 fails.

All observed oscillations occur through Hopf bifurcation at a single fixed point (Unstable Spiral). No oscillation via a global three-fixed-point mechanism is possible. Because the sigmoid function is flat at both ends $(F' \approx 0)$, the outer fixed points in any multiple-point system will always have a negative trace $(\tau < 0)$. Therefore, the emergence of a Saddle point $(\Delta < 0)$ mathematically guarantees the existence of Stable Nodes/Spirals. These stable points act as inescapable attractors and completely prevent the formation of any limit cycle.

w_{EE}	w_{II}	I_E^{ext}	C1	C2	C3	Fixed Point State
11	0	0.50	1	1	1	**Unstable Spiral** (Oscillation)
11	5	0.00	0	0	0	Stable Spiral
11	5	0.00	1	0	0	Saddle Point
11	5	0.00	0	0	0	Stable Node
14	2	0.00	0	0	1	Stable Spiral
14	2	0.00	1	0	1	Saddle Point $(\Delta < 0)$
5	0	1.00	0	1	1	Stable Spiral

Table 4: Representative numerical evaluations of the analytical conditions[5].

3.2.1 Application to the parameters

For the NMA default parameters ($w_{EE} = 9$, $w_{EI} = 4$, $w_{IE} = 13$, $w_{II} = 11$):

$$w_{EI}w_{IE} = 52 < 99 = w_{EE}w_{II} \tag{24}$$

Condition 3 is violated: the cross-term is $-47F'_EF'_I$, strongly competitive. The trace and determinant conditions cannot be simultaneously satisfied at any fixed point. Numerically, the saddle point achieves F'_E up to 0.28 with a positive trace, but $\Delta < 0$; the stable spiral has $\Delta > 0$, but $F'_E \leq 0.12$ and a negative trace. No Hopf bifurcation occurs regardless of I_E^{ext}.

For Li et al. ($w_{EE} = 16$, $w_{EI} = 26$, $w_{IE} = 20$, $w_{II} = 1$, $\theta_E = 5.0$): $w_{EI}w_{IE} = 520 \gg 16 = w_{EE}w_{II}$ (Condition 3), $u_E = 5.24 \approx \theta_E = 5.0$ (Condition 2), and $w_{EE}F'_E = 4.7$ (Condition 1).

[5]The system only oscillates when C1 (Trace lower bound), C2 (Determinant upper bound), and C3 (Cross-term inequality) are simultaneously satisfied. Note the emergence of multi-fixed-point regimes (Stable-Saddle-Stable) when C2 or C3 is violated.

All three conditions are satisfied, and the system exhibits a Hopf bifurcation at a single Unstable Spiral.

3.3 Further discussion

Our analytical conditions are consistent with the numerical bifurcation results of Li et al. (2022), who investigated the role of self-feedback loops in regulating gamma oscillations in the Wilson-Cowan model. Their one-parameter bifurcation with respect to w_{II} (their Figure 3A) shows that increasing inhibitory self-feedback suppresses oscillation. This directly follows from our Condition 3: since the cross-term product $w_{EI}w_{IE}$ is fixed, increasing w_{II} raises $w_{EE}w_{II}$ and eventually breaks the inequality $w_{EI}w_{IE} > w_{EE}w_{II}$.

For excitatory self-feedback, their one-parameter bifurcation with respect to w_{EE} (their Figure 6A) reveals that oscillation exists only within a limited window ($13.57 < w_{EE} < 35$). Their study concludes that "the excitatory self-feedback loop promotes the generation of gamma oscillations". However, this statement only explains the lower boundary of the window. Our mathematical analysis clarifies exactly why both boundaries exist:

- The **lower bound** reflects Condition 1: the system needs enough excitatory gain to overcome decay and make the trace positive ($\tau > 0$).
- The **upper bound** reflects Condition 2: a very large w_{EE} either forces the determinant below zero ($\Delta < 0$, creating a Saddle point) or pushes the system into deep saturation ($F'_E \approx 0$, making $\tau < 0$).

Their transfer function analysis (their Eq. 13–14) assumes $w_{EE}K_E < 1$, which mathematically requires the system to stay safely within this exact intermediate window.

We refer to the following papers:

Inhibitory self-feedback. Li et al. (2022):

> The present results reveal that, on one hand, the inhibitory self-feedback loop is not conducive to the generation of gamma oscillations, and increased inhibitory self-feedback strength facilitates the enhancement of the oscillation frequency. On the other hand, the excitatory self-feedback loop promotes the generation of gamma oscillations […] Inhibitory and excitatory self-feedback loops play a complementary role in generating and regulating the gamma oscillation in Wilson-Cowan model.

Time constants. Keeley et al. (2019):

> The models' flexibility to capture the broad range of gamma behavior depends directly on the timescales that represent recruitment of the excitatory and inhibitory firing rates. In particular, we find that weakly modulated gamma oscillations occur robustly when the recruitment timescale of inhibition is faster than that of excitation.

4 Key References

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6. **Li, D., Liu, S. & Wang, J. (2022).** Bidirectionally regulating gamma oscillations in Wilson-Cowan model by self-feedback loops: a computational study. *Frontiers in Systems Neuroscience*, 16, 723237. doi: 10.3389/fnsys.2022.723237

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